

Homework #2

Zeynep Noz Öziürk

22403317

① ④ $\frac{dy}{dx} = (y^2 - 1) \cdot e^x$

$$\frac{dy}{y^2 - 1} = e^x dx$$

Integrate both sides:

$$\frac{1}{2} \int \frac{1}{y-1} - \frac{1}{y+1} dy = \int e^x dx$$

$$\ln \left| \frac{y-1}{y+1} \right| = 2e^x + C, \text{ for } C \in \mathbb{R}$$

$$\frac{y-1}{y+1} = e^{2e^x + C}, \text{ for } B \in \mathbb{R}$$

$$\text{for } y(0) = 3; \quad B = \frac{1}{2} e^{-2}$$

$$\frac{y-1}{y+1} = \frac{1}{2} e^{2e^x - 2}$$

$$y-1 = \frac{1}{2} y e^{2e^x - 2} + \frac{1}{2} e^{2e^x - 2}$$

$$y - \frac{1}{2} e^{2e^x - 2} = \frac{1}{2} e^{2e^x - 2} + 1$$

$$y(x) = \frac{\frac{1}{2} e^{2e^x - 2} + 1}{1 - \frac{1}{2} e^{2e^x - 2}} //$$

⑤ $\frac{dy}{dx} = y \ln y$

$$\frac{dy}{y \ln y} = dx$$

Integrate both sides:

$$\int \frac{dy}{y \ln y} = \int dx \Rightarrow \ln |\ln y| = x + C, \text{ for } C \in \mathbb{R}$$

$$\text{for } u = \ln y, \quad du = \frac{1}{y}$$

$$\ln |\ln y| = x + C$$

$$|\ln y| = B \cdot e^x, \text{ for } B \in \mathbb{R} \Rightarrow y = e^{B \cdot e^x}$$

$$\text{for } y(0) = 2; \quad B = \ln 2$$

$$y(x) = e^{e^x \ln 2} //$$

② ④ $x^2 dx + 2y dy = 0$

$$\frac{\partial}{\partial y} x^2 = \frac{\partial}{\partial x} 2y = 0 \quad (\text{exactness condition})$$

$$f(x, y) = \frac{x^3}{3} + y^2 = C, \text{ for } C \in \mathbb{R}$$

$$\text{General Solution: } y = \pm \sqrt{C - \frac{x^3}{3}}$$

$$F(1, 10) = \frac{1}{3} + 100 = \frac{301}{3} \quad (C)$$

$$y(x) = \pm \sqrt{\frac{301}{3} - \frac{x^3}{3}} //$$

⑤ $(\sin y + y \cos x) dx + (\sin x + x \cos y) dy = 0$

$$\frac{\partial}{\partial y} (\sin y + y \cos x) = \cos y + \cos x = \frac{\partial}{\partial x} (\sin x + x \cos y) \quad (\text{exactness condition})$$

$$f(x, y) = x \cdot \sin y + y \cdot \sin x = C, \quad C \in \mathbb{R}$$

$$f(2, 3) = 2 \cdot \sin 3 + 3 \cdot \sin 2 \approx 3 = C$$

$$x \cdot \sin y + y \cdot \sin x - 3 = 0 //$$

③ ④ $(ye^{-x} + 1) dx + (xe^{-x}) dy = 0$

$$\frac{\partial}{\partial y} (ye^{-x} + 1) = e^{-x} + \frac{\partial}{\partial x} (xe^{-x}) \quad (\text{exactness condition does not hold})$$

$$\text{Let } \sigma(ye^{-x} + 1) dx + \sigma(xe^{-x}) dy = 0$$

$$\frac{\partial}{\partial y} (\sigma(ye^{-x} + 1)) = \sigma_y (ye^{-x} + 1) + \sigma \cdot e^{-x} = 1$$

$$\frac{\partial}{\partial x} (\sigma(xe^{-x})) = \sigma_x (xe^{-x}) + \sigma (e^{-x} - xe^{-x})$$

$$\text{Let } \sigma_y = 0 \Rightarrow \sigma = \sigma(x);$$

$$\sigma e^{-x} = \sigma' (xe^{-x}) + \sigma (e^{-x} - xe^{-x})$$

$$\sigma(x) = e^x$$

$$(y + e^x) dx + x dy = 0$$

$$f(x, y) = xy + e^x = C, \text{ for } C \in \mathbb{R}$$

$$y(x) = \frac{C - e^x}{x} //$$

⑤ $\frac{dy}{dx} = \frac{e^{-2y}}{\tan x}$

$$\int e^{-2y} dy = \int \frac{\cos x}{\sin x} dx \Rightarrow \int e^{-2y} dy = \int \frac{du}{u}$$

$$\text{for } u = \sin x \quad du = \cos x \cdot dx$$

$$e^{-2y} = -2 \ln |\sin x| + C, \text{ for } C \in \mathbb{R}$$

③ $e^{2y} dx - \tan x \cdot dy = 0$

$$\frac{\partial}{\partial y} (\sigma e^{2y}) = \frac{\partial}{\partial x} (-\sigma \tan x)$$

$$\sigma_y e^{2y} + 2\sigma e^{2y} = -\sigma_x \tan x - \sigma \sec^2 x$$

$$\text{For } \sigma = \sigma(x); \quad \frac{\sigma'}{\sigma} = \frac{2e^{2y} + \sec^2 x}{-\tan x} \neq f(x) \text{ or } f(y)$$

$$\text{For } \sigma = \sigma(y); \quad \frac{\sigma'}{\sigma} = \frac{-\sec^2 x - 2e^{2y}}{e^{2y}} \neq f(x) \text{ or } f(y)$$

$$\text{Let } \sigma(x, y) = \frac{e^{-2y}}{\tan x} \text{ by inspection.}$$

$$\frac{\partial}{\partial y} (\sigma e^{2y}) = 0 \text{ and } \frac{\partial}{\partial x} (\sigma(-\tan x)) = 0 \quad (\text{exactness condition})$$

$$\frac{1}{\tan x} \cdot dx - e^{-2y} dy = 0$$

$$\int \frac{dx}{\tan x} = \ln |\sin x| + A(y) + C, \text{ for } C \in \mathbb{R}$$

$$\int -e^{-2y} dy = \frac{1}{2} e^{-2y} = A(y)$$

$$F(x, y) = \ln |\sin x| + \frac{1}{2} e^{-2y} + C = B, \text{ for } B \in \mathbb{R}$$

$$\ln |\sin x| + \frac{1}{2} e^{-2y} = A, \text{ for } A \in \mathbb{R} //$$